# TRIBHUVAN UNIVERSITY
# INSTITUTE OF ENGINEERING

PURWANCHAL CAMPUS
DHARAN, SUNSARI

Major Project Proposal/Report on

**Design, Modeling, and Kinematic-Dynamic Analysis of a Stick Counting and Packaging System**

BY

SEWAK SUNAR - PUR080BME051
PRIT NARAYAN YADAV - PUR080BME063
INDRA NARAYAN SHA - PUR080BME062
SASHI MANDAL - PUR080BME049

TO

DEPARTMENT OF MECHANICAL ENGINEERING
11th June, 2026

## ACKNOWLEDGEMENT

We are also deeply grateful to the local production units and workshops we visited during our initial field research. The opportunity to document real-world operational challenges and design-related problems first-hand was vital in identifying the specific research gaps that this modular stick-packaging system aims to address.

Furthermore, we acknowledge the collaborative work of our team members, whose shared technical competencies and dedication have been central to this systematic design approach. This project has provided a valuable platform to bridge the gap between theoretical knowledge and professional engineering practice.

Finally, we would like to thank our institution for providing the environment to develop the technical competencies and foundational skills required to initiate the conceptual design process for the *Design, Modeling, and Kinematic-Dynamic Analysis of a Stick Counting and Packaging System*.

# ABSTRACT

Mechanical automation plays a vital role in improving the efficiency and reliability of repetitive industrial operations. This project focuses on the **design, modeling, and kinematic-dynamic analysis** of a semi-automatic stick counting and packaging mechanism. The primary objective is to study and evaluate the performance of various mechanical mechanisms that can be integrated to achieve accurate stick counting, bundle formation, and packaging operations.

The proposed system consists of a stick feeding mechanism, a purely mechanical counting mechanism, an intermittent motion transfer mechanism, and a bundle holding and packaging mechanism. The study emphasizes the application of machine design principles, including linkages, ratchet-pawl systems, cam mechanisms, and intermittent motion mechanisms, to achieve synchronized and repeatable operations without relying on complex electronic control systems.

The project methodology involves **conceptual design, engineering calculations, 3D modeling using FreeCAD, and kinematic and dynamic analysis** to investigate the motion characteristics, force transmission, and torque requirements of the proposed mechanisms. The interaction between the individual mechanisms and their overall contribution to system performance are analyzed to identify an efficient and practical design configuration. The results of the analysis will provide insights into the feasibility and effectiveness of the proposed design, as well as potential areas for improvement and optimization in future iterations. This project aims to contribute to the field of mechanical automation by demonstrating a practical application of machine design principles in a real-world context.

# TABLE OF CONTENTS

# LIST OF FIGURES

# LIST OF TABLES

# LIST OF ABBREVIATION

**CAD** Computer Aided Design

**CAM** Computer Aided Manufacturing

**FEA** Finite Element Analysis

**BOM** Bill of Materials

**CNC** Computer Numerical Control

**RPM** Revolutions Per Minute

**DOF** Degrees of Freedom

**FOS** Factor of Safety

**ROI** Return on Investment

**MTBF** Mean Time Between Failures

**MTTR** Mean Time To Repair

# CHAPTER 1: INTRODUCTION

## 1.1 Background Theory

Mechanical automation has become an essential aspect of modern manufacturing, where repetitive tasks are accomplished through the integration of efficient mechanisms. Among these tasks, counting and packaging operations require accurate motion control, synchronization, and repeatability. Traditionally, such operations are performed manually or with electronically controlled systems, which may increase labor requirements, system complexity, and cost.

This project focuses on the study and analysis of mechanical mechanisms that can be integrated to perform semi-automatic stick counting and packaging operations. The system utilizes purely mechanical components to achieve controlled and sequential motion without relying on complex electronic sensors or controllers.

The theoretical foundation of the project is based on several core areas of mechanical engineering:

- Kinematics of Mechanisms: The study of displacement, velocity, and acceleration of moving machine elements such as linkages, cams, ratchet-pawl systems, and intermittent motion mechanisms.
- Dynamics of Mechanisms: Investigation of the forces, torques, and power requirements necessary to drive the mechanical system while ensuring smooth and stable operation.
- Machine Element Design: Application of engineering principles for the design of shafts, gears, bearings, springs, and other machine components to achieve reliable performance.
- Structural Mechanics and Material Selection: Evaluation of stresses and selection of suitable materials to ensure adequate strength, durability, and operational safety.
- Computer-Aided Design and Analysis: Three-dimensional modeling and assembly development using FreeCAD, followed by kinematic and dynamic analysis to validate the proposed mechanisms and their interactions.

## 1.2 Problem Statement

Many small and medium-scale manufacturing industries continue to rely on manual methods for counting and packaging stick-type products. Manual operations are repetitive, time-consuming, and prone to inconsistencies in counting accuracy and packaging quality. Although automated systems are available, they often depend on electronic sensors and complex control systems that increase both manufacturing and maintenance costs.

There been currenlty used the drum type stick counter, which is a purely mechanical device that counts sticks as they pass through a rotating drum equipped with counting mechanisms. While this method is simple and does not require electronic components, it has limitations in terms of speed, accuracy, and adaptability to different stick sizes and packaging requirements. The need for a more efficient and versatile solution has led to the exploration of various mechanical mechanisms that can be integrated to achieve semi-automatic stick counting and packaging operations.

From a mechanical engineering perspective, there is a need to study and develop efficient mechanical mechanisms capable of performing these operations through coordinated motion and force transmission. However, the selection, integration, and validation of such mechanisms require detailed kinematic and dynamic investigation.

This project addresses this need by studying and analyzing various mechanical mechanisms, including linkage systems, ratchet-pawl mechanisms, cam mechanisms, and intermittent motion mechanisms, for their application in a semi-automatic stick counting and packaging system. The primary focus is to understand the behavior, motion characteristics, and feasibility of these mechanisms through engineering calculations, CAD modeling, and kinematic-dynamic analysis.

The ultimate goal is to develop a validated mechanical model that demonstrates the potential for low-cost, efficient automation in stick counting and packaging operations, providing a foundation for future prototype development and further research in this area.

# CHAPTER 2: LITERATURE REVIEW

This chapter reviews key literature on mechanical counting and bundling systems, emphasizing intermittent-motion mechanisms, design methods, and practical constraints for low-cost, high-speed and compact size solutions.

## 2.1 Mechanical Indexing and Counting Systems

Mechanical indexing devices (ratchet-pawl assemblies, Geneva mechanisms, escapements) and cam-driven linkages have long been used for precise, repeatable motion in packaging equipment. Their deterministic timing and robustness make them attractive where electronic control is impractical or costly.

## 2.2 Design Methodologies and Simulation

Modern machine design combines classical kinematics and dynamics with CAD-based modeling and simulation. Kinematic analysis defines motion profiles and timing; FEA and simple dynamic models inform structural sizing, while rapid prototyping shortens iteration cycles.

## 2.3 Materials and Manufacturing Considerations

Selection of materials, surface finishes, and tolerances affects wear, friction, and reliability. For small-scale, low-cost machines, a balance is needed between durable components (hardened pins, low-friction bushes) and economical manufacturing methods (sheet metal, simple machining).

## 2.4 Existing Low-Cost Solutions and Gaps

Existing literature describes semi-automatic incense bundling and counting solutions that use simple fixtures and manual feeds; however, comprehensive studies on fully mechanical, low-cost counting-and-binding machines tailored to small manufacturers are scarce. This gap motivates the present work.

## Summary

This chapter has reviewed relevant literature on mechanical counting and bundling systems, design methodologies, and material considerations. The insights gained will inform the conceptual design and analysis of a compact, low-cost stick counting and packaging mechanism in subsequent chapters.

# CHAPTER 3: RELATED THEORY

This chapter presents the theoretical background that informs the design choices for a mechanical incense stick counting and binding machine. Emphasis is placed on kinematics, actuation, material behaviour, and practical feeding/counting strategies.

## 3.1 Kinematic Principles of Linkages and Mechanisms

Understanding the kinematics of linkages (four-bar, slider-crank) and mechanisms (Geneva, ratchet-pawl) is essential for designing the motion paths and timing relationships required for stick feeding, counting, and bundling. The synthesis of these mechanisms must ensure smooth, repeatable motion while minimizing impact forces and wear.

## 3.2 Cam and Follower Design

Cam design translates continuous rotary input into prescribed follower motion. Choosing appropriate lift profiles and follower types (roller, flat-faced) reduces impact loading and wear—important when handling slender, fragile sticks.

## 3.3 Force, Torque and Power Estimation

Sizing power transmission elements requires estimating steady and peak loads from friction, impact during transfers, and binding actions. Simple static and dynamic calculations, augmented by safety factors, guide selection of input drive, gear ratios, and bearings.

## 3.4 Materials, Wear and Tolerance Management

Component materials and surface treatments influence longevity and reliability. For low-cost manufacture, combining wear-resistant pins and low-friction polymer guides often yields good trade-offs. Specifying realistic tolerances and clearances mitigates jamming and increases reproducibility.

## 3.5 Counting, Feeding and Binding Principles

Reliable counting is achieved via mechanical escapements, cam-actuated gates, or indexed pockets that present one item per cycle. Passive guides and compliant elements (springs, rollers) reduce misfeeds. Binding methods (adhesive, mechanical clamps, twisting) must be chosen for simplicity, speed, and maintainability.

## 3.6 Validation and Testing Approach

Prototype validation focuses on cycle-count accuracy, throughput stability, and wear patterns. Measurements using tachometers, run-count logs, and periodic inspection help quantify performance and guide iterative improvements.

## Summary

This chapter has outlined the key theoretical concepts that underpin the design of a mechanical stick counting and bunding machine. By applying principles of kinematics, cam design, force estimation, material selection, and feeding strategies, we can develop a robust, efficient mechanism that meets the project objectives. The next chapter will explore the feasibility of various design concepts through preliminary calculations and prototyping.

# CHAPTER 4: FEASIBILITY STUDY

## 4.1 Technical Feasibility

Even we will be just desinging the mechanism, still we will be study the the manufacturing process. The design will be based on standard mechanical components and manufacturing techniques to ensure that the proposed stick counting and packaging system can be feasibly produced. The components will be selected for their availability, cost-effectiveness, and suitability for the intended application. The design will also consider ease of assembly and maintenance to enhance the overall feasibility of the system.

Table 4.1: Components Specifications

| Component | Typical Specification |
|---|---|
| Frame | Mild steel sheet / welded frame |
| Indexing mechanism | Geneva or cam-driven with hardened pins |
| Bearings | Standard ball or sleeve bearings (sizes as per shaft) |
| Feed guides | Low-friction polymer or brass inserts |
| Drive | Hand-crank or small geared motor (10-50 W) |
| Binding mechanism | Simple clamp or adhesive dispenser |
| Fasteners | Standard bolts, nuts, washers |

## 4.2 Modeling and Simulation Software Feasibility

Modeling and simulation are crucial aspects of the design process. We use the FreeCAD software for 3D modeling and kinematic analysis. FreeCAD is an open-source parametric CAD software that supports the design of mechanical components and assemblies. It provides tools for creating detailed 3D models, performing simulations, and analyzing the motion of mechanisms. The software's capabilities align well with the requirements of our project, allowing us to effectively model the stick counting and packaging system, analyze its kinematics and dynamics, and optimize the design for performance and reliability. Also for the purpose of dynamic analysis, we will be using the Python programming language with libraries.

For managing the variables, dimensions we will be using Excel spreadsheets during the design process. This will allow us to systematically organize and analyze the data related to the design parameters, material properties, and performance metrics of the proposed system. The use of Excel will facilitate efficient data management and support informed decision-making throughout the design and analysis phases of the project.

## 4.3 Economic Feasibility

We will go insight into the cost of materials, manufacturing, and maintenance of the proposed stick counting and packaging system. The design will prioritize cost-effectiveness by utilizing standard components and materials that are readily available in the market. The manufacturing process will be streamlined to minimize labor and production costs, while still ensuring the quality and reliability of the final product. Additionally, the system will be designed for easy maintenance to reduce long-term operational costs. The economic feasibility will be evaluated by comparing the estimated costs of the proposed system with the potential benefits it offers in terms of improved efficiency and productivity in stick counting and packaging operations.

## 4.4 Operational Feasibility

Operational feasibility will be assessed by evaluating the ease of use, reliability, and maintenance requirements of the proposed stick counting and packaging system. The design will focus on creating a user-friendly interface for operators, ensuring that the system can be easily operated with minimal training. Reliability will be a key consideration, with the design incorporating robust components and mechanisms to minimize downtime and ensure consistent performance. Maintenance requirements will also be addressed by designing the system for easy access to critical components, allowing for quick repairs and routine maintenance tasks. The operational feasibility will be further evaluated through testing and feedback from potential users to ensure that the system meets their needs and expectations in a real-world setting.

# CHAPTER 5: METHODOLOGY

## 5.1 System Overview

The system comprises an input hopper, feed guides, an indexing mechanism (cam or Geneva), a counting gate, a binding station, and an output collection tray. Drive power is provided by a hand crank or small motor coupled through gears to achieve the required torque and speed.

## 5.2 Sequence of Operations

A block diagram represents the functional flow (Fig. 5.1a). Each block is coordinated by the indexing cycle to ensure one-to-one correspondence between counts and bindings.

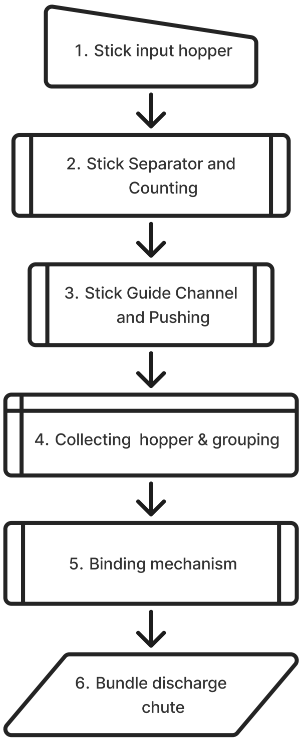
Iterative adjustments will be made based on test results to optimize performance and reliability. Flow charts (Fig. 5.1b) will be used to validate the sequence of operations and identify potential failure points for mitigation.

## 5.3 Overall Design and Analysis Workflow

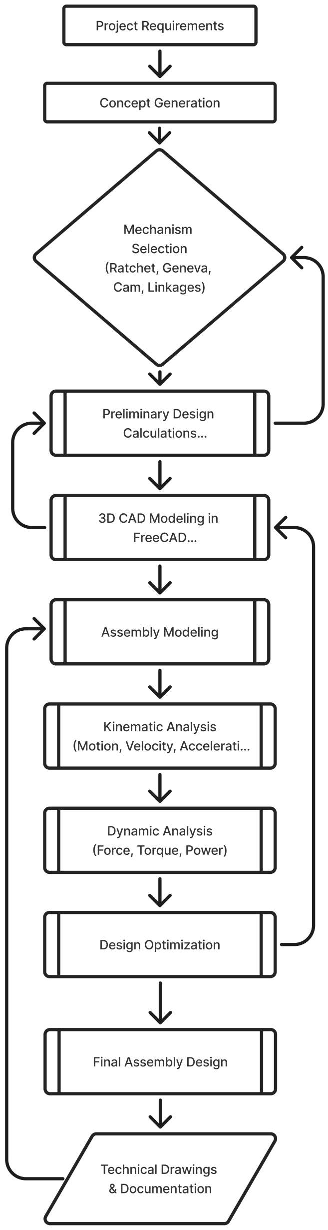
Design and analysis will begin with concept generation and selection, followed by detailed kinematic design of the indexing mechanism. CAD modeling will be used to refine the design, followed by kinematic and dynamic analysis and finally testing and documentation.
Iterative adjustments will be made based on test results to optimize performance and reliability. Flow charts (Fig. 5.1b) will be used to validate the sequence of operations and identify potential failure points for mitigation.

## 5.4 CAD Modeling and Simulation

CAD software (SolidWorks) will be used for 3D modeling of the mechanism, allowing for detailed design and interference checks. Kinematic analysis will be performed to ensure the desired motion profiles are achieved, while dynamic analysis will evaluate forces and stresses to inform material selection and structural design. Simulation results will be validated against theoretical calculations to ensure accuracy and reliability of the design before prototyping. Flow chart 5.2 shows the workflow.




(a) Functional block diagram of the mechanical counting and bundling machine.




(b) Flow chart depicting the sequence of operations and decision points for the mechanical counting and bundling machine.

Figure 5.1: System block diagram and process flowchart.

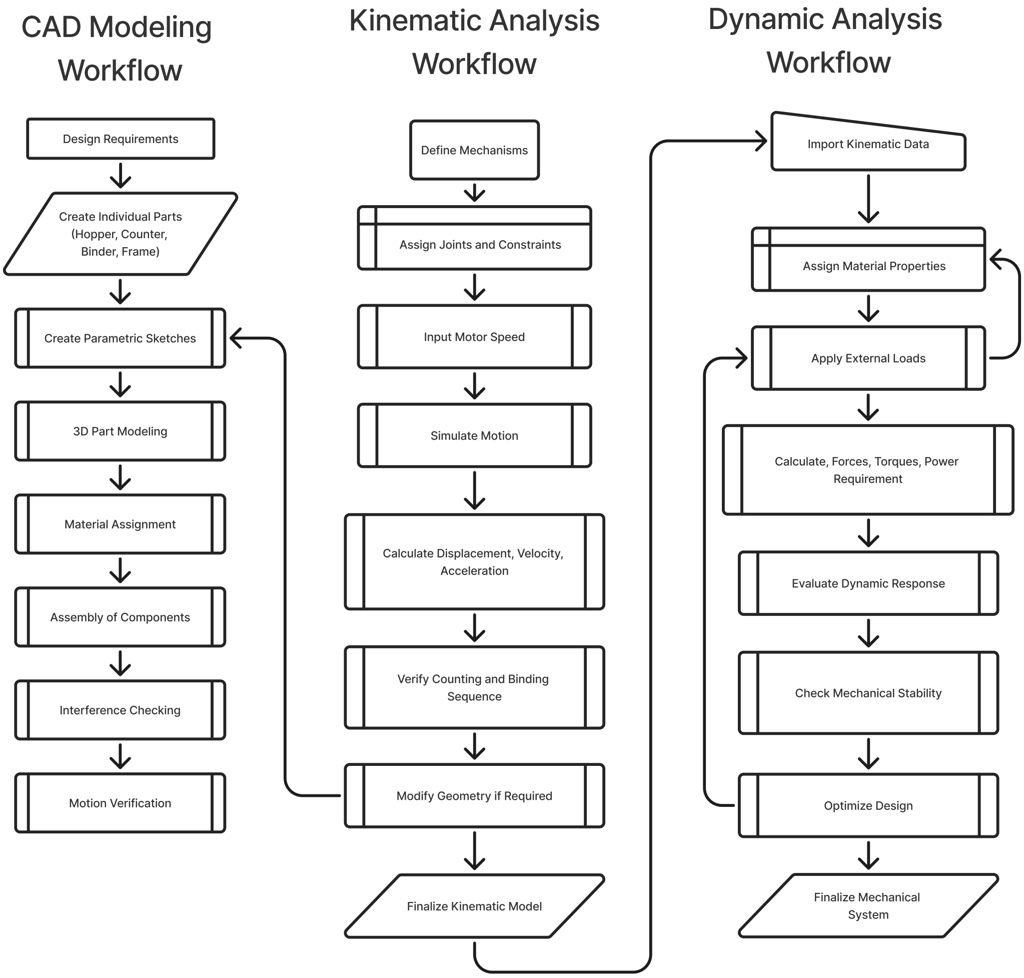



Figure 5.2: Flow chart illustrating the CAD modeling and simulation workflow for the mechanical counting and bundling machine.

CHAPTER 6: RESULT AND ANALYSIS

## 6.1 Expected Output

The prototype is expected to deliver grouped sets of incense sticks at a predictable count per cycle with low variance. Initial throughput targets are 30-120 groups per minute depending on drive method and operator skill.

## 6.2 Performance Metrics

Key metrics to evaluate the design include:

- Counting accuracy (items per group vs target)
- Throughput (groups per minute)
- Mean time between jams (MTBJ)
- Wear indicators (visual inspection of guides, pins)
- Energy input per cycle (if motorized)

## 6.3 Success Criteria

The design will be considered successful if it achieves $\geq$ 95% counting accuracy, meets a practical throughput for small-scale production, and can be fabricated and maintained at a low cost.

## 6.4 Future Work

Future improvements may include simplifying the binding mechanism, adding lightweight sensing for automated reject handling, and optimizing cam profiles for smoother operation.

# Minor Project

**Project lead**

| | |
|---|---|
| Sewak Sunar | Prit Narayan |
| Indra Narayan | Sashi Mandal |

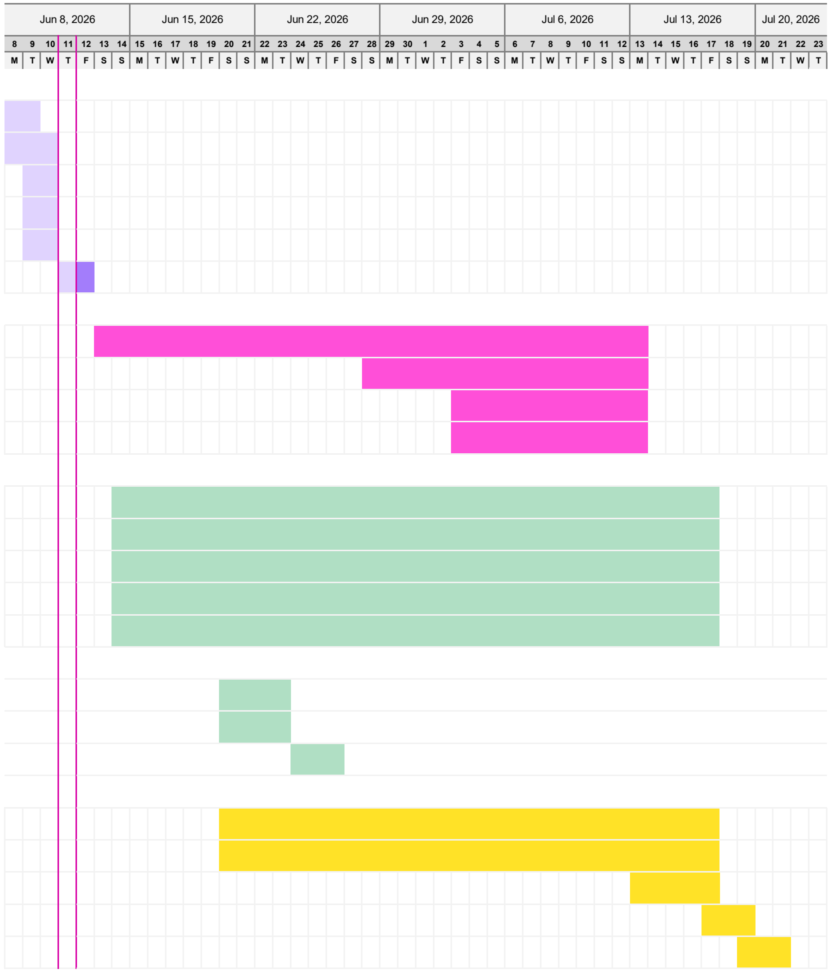
**Project start:** **Mon, 6/8/2026**

**Display week:** **1**

## TeamStick

GANTT CHART
[https://github.com/sewaksunar/auto-stick-packer](https://github.com/sewaksunar/auto-stick-packer)

| TASK | ASSIGNED TO | PROGRESS | START | END |
|---|---|---|---|---|
| **Proposal & Conceptual Design** | | | | |
| Team Formation & Title Selection | Team | 100% | 6/8/26 | 6/9/26 |
| Concept Generation & Preliminary Design | Sewak Sunar | 100% | 6/8/26 | 6/10/26 |
| Literature Review & Existing Solution Analysis | Prit Narayan | 100% | 6/9/26 | 6/10/26 |
| Technical and Economic Feasibility Study | Sashi Mandal | 100% | 6/9/26 | 6/10/26 |
| Problem Identification & Requirement Analysis | Indra Narayan | 100% | 6/9/26 | 6/10/26 |
| Title Declaration & Proposal Defense | Team | 50% | 6/11/26 | 6/12/26 |
| **Detailed Modeling (FreeCAD) & Tool Validation** | | | | |
| Detailed Modeling (FreeCAD) & Tool Validation | Sewak Sunar | 0% | 6/13/26 | 7/13/26 |
| FreeCAD Assembly | Sewak Sunar | 0% | 6/28/26 | 7/13/26 |
| Kinematic Analysis | Sewak Sunar | 0% | 7/3/26 | 7/13/26 |
| Dynamic Analysis | Sewak Sunar | 0% | 7/3/26 | 7/13/26 |
| **Manufacturing Aspect & Feasibility** | | | | |
| Material Selection | Prit Narayan | 0% | 6/14/26 | 7/17/26 |
| Tool Selection | Prit Narayan | 0% | 6/14/26 | 7/17/26 |
| Cost Estimation | Sashi Mandal | 0% | 6/14/26 | 7/17/26 |
| Production Feasibility | Indra Narayan | 0% | 6/14/26 | 7/17/26 |
| Testing and validation | Team | 0% | 6/14/26 | 7/17/26 |
| **Minor Project Title Defence (Mid-Term Presentation)** | | | | |
| Preparation of Report on Process | Sewak Sunar | 0% | 6/20/26 | 6/23/26 |
| Prepation of Presentation on Process | Sewak Sunar | 0% | 6/20/26 | 6/23/26 |
| Minor Project Title Defence (Mid-Term Presentation) | Team | 0% | 6/24/26 | 6/26/26 |
| **Documentation & Defense** | | | | |
| Final Design Refinement & Standards Validation | Team | 0% | 6/20/26 | 7/17/26 |
| Report Writing (APA 7th Edition Standards) | Sewak Sunar | 0% | 6/20/26 | 7/17/26 |
| Evaluate progress | Team | 0% | 7/13/26 | 7/17/26 |
| Milestone: Final Report Submission | Team | 0% | 7/17/26 | 7/19/26 |
| Milestone: Final Presentation & Defense | Team | 0% | 7/19/26 | 7/21/26 |



# BIBLIOGRAPHY

[1] John J. Uicker, Jr., Gordon R. Pennock, and Joseph E. Shigley, *Mechanical Engineering Design*, 9th ed., McGraw-Hill, 2011.

[2] John J. Uicker, Jr., Gordon R. Pennock, and Joseph E. Shigley, *Theory of Machines and Mechanism*, 5th ed., Oxford University Press, 2017.

[3] Proceedings of the 4th International Conference on Mechanical Design and Simulation. (2024). *Mechanical Design and Simulation: Exploring Innovations for the Future*. Singapore: Springer. https://doi.org/10.1007/978-981-97-7887-4

[4] IndiaMART. (2023). *Stainless Steel incense stick machine die, production capacity: 15-20 kg/hr, 150-200 strokes/min - PDF catalogue*. Retrieved from `https://pdf.indiamart.com/impdf/2850397300948/25812023/incense-stick-making-machine-die.pdf`